FULL CARD 30 — Climate, Energy, Water and Evidence

INTEGRATED • SOCIAL STUDIES • CHEMISTRY • PHYSICS • MATHEMATICS

INVESTIGATION • A school compares four months of rainfall, average temperature, solar electricity generation and water use. The table is illustrative, not a real weather record.

Month	Rainfall (mm)	Temp (°C)	Solar output (kWh)	Water use (kL)
Jan	20	22	420	310
Mar	15	28	510	335
May	30	34	560	390
Jul	240	29	390	360
Sep	160	28	430	345

Mathematics • calculate and compare

Rainfall range = 240 – 15 = 225 mm. Mean rainfall = (20 + 15 + 30 + 240 + 160) ÷ 5 = 465 ÷ 5 = 93 mm. Highest listed solar output occurs in May, but the table alone cannot identify all causes of the differences.

Physics • energy transfer

Solar panels convert some incoming light energy into electrical energy. Output can depend on sunlight, panel orientation, shading, temperature, system efficiency, and equipment condition. More sunshine does not always mean proportionally more electricity.

Chemistry • water and emissions

Water treatment may involve filtration, sedimentation, and disinfection. Combustion of carbon-containing fuels produces carbon dioxide; some fuels can also release pollutants such as sulfur dioxide or nitrogen oxides. The exact emissions depend on fuel and technology.

Social Studies • resource choices

Water use reflects climate, population, farming, industry, habits, infrastructure, and pricing. A fair policy considers access, affordability, public health, and ecological needs—not just the total volume consumed.

Evidence quality checklist

Question	Why it matters
Are months comparable?	Season and daylight affect patterns.
Are units consistent?	Mixed units can create false comparisons.
Are key variables missing?	Population, rainfall timing and equipment state may matter.
Does correlation prove cause?	No; multiple explanations may fit the same pattern.

Extended response: Recommend one action to reduce water use and one action to improve clean-energy reliability. For each, state the expected benefit, a possible trade-off, and data needed to evaluate success.